

Predictive Modeling Plan – Geldium Delinquency Risk Assessment

Step 1: Predictive Model Logic

Selected Model

The recommended model for predicting customer delinquency is Logistic Regression, supported by Decision Tree analysis for additional explainability. Logistic regression is highly suitable for binary classification problems such as predicting whether a customer will become delinquent or not. Decision trees can further help visualize risk segmentation and explain prediction logic to business stakeholders.

Predictive Modeling Workflow

1. Collect and preprocess customer financial data.
2. Handle missing values using statistical imputation techniques.
3. Select relevant risk-related features from the dataset.
4. Split the dataset into training and testing datasets.
5. Train the logistic regression model using historical customer behavior.
6. Generate delinquency probability scores for customers.
7. Classify customers into low-risk or high-risk categories based on prediction thresholds.
8. Evaluate model performance using accuracy, recall, F1-score, ROC-AUC, and fairness metrics.

Top 5 Input Features

- Payment history
- Credit utilization ratio
- Debt-to-income ratio
- Outstanding balance
- Recent credit activity

Step 2: Model Justification

Logistic regression was selected because it is highly interpretable, efficient, and widely used in financial risk modeling for binary classification problems. The model provides probability-based outputs that help Geldium identify customers with elevated delinquency risk while maintaining transparency for regulatory compliance and business decision-making.

Compared to more complex models such as neural networks, logistic regression offers easier explainability and simpler deployment while still delivering reliable performance on structured financial datasets. Decision trees can additionally support explainability by clearly identifying the factors contributing to high-risk classifications. This combination aligns well with Geldium's need for responsible, explainable, and scalable AI-driven risk assessment.

Step 3: Evaluation Strategy

Model Performance Metrics

- Accuracy: Measures overall prediction correctness.
- Precision: Evaluates how many predicted delinquent customers are actually delinquent.
- Recall: Measures how effectively the model identifies actual delinquent customers.
- F1 Score: Balances precision and recall to reduce prediction errors.
- ROC-AUC Score: Measures how effectively the model distinguishes between delinquent and non-delinquent customers.
- Confusion Matrix: Helps analyze false positives and false negatives.

Fairness and Bias Assessment

The model will undergo fairness testing to ensure no customer groups are unfairly disadvantaged. Bias checks will include reviewing prediction outcomes across demographic groups and monitoring for proxy bias in sensitive variables. Explainability techniques such as decision trees and feature importance analysis will help ensure transparency in decision-making.

Model Improvement Strategy

If performance is weak, improvements may include:

- Feature engineering
- Rebalancing delinquency classes
- Hyperparameter tuning
- Comparing alternative model approaches
- Additional fairness testing and validation

Conclusion

The proposed predictive modeling approach combines explainability, fairness, and predictive effectiveness to support Geldium's delinquency risk assessment goals. Logistic regression provides a transparent and regulator-friendly solution for predicting customer delinquency, while fairness checks and explainability measures help ensure responsible AI adoption. This framework will support proactive intervention strategies, improved collections prioritization, and more reliable financial risk management.